

CPSC 483: Introduction to Machine Learning

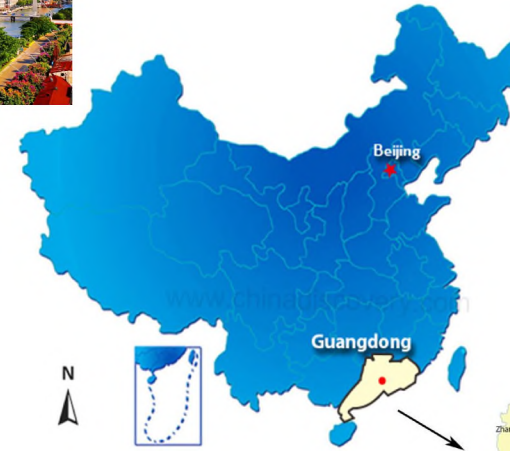
Instructor: Anli Ji

Fall 2026

Cal State Fullerton

A little about me..

- Assistant Professor at CSUF
- From Guangdong, China
- Lived in Georgia for 9 years
 - Ph.D. in Georgia State University
- Research Interests
 - Time Series Analysis (i.e., Multivariate)
 - Data Mining
 - Machine-learning
- If you must know, I like anime and board games :D



Guangdong China Map



A little about me..

Contact

- Email: anli.ji@fullerton.edu (best way to reach me)
 - Please include course title "CPSC 483" (to make sure I do not disregard your email)
 - Please send email from your **fullerton.edu** account (no reply to any free web-based email accounts such as Gmail, Hotmail or Yahoo)
- Office: CS-538
- Office hour: Tuesday and Thursday 9:30 am – 11:00 am or by appointment

Learning Management System (LMS): Canvas

- Student help desk: (657) 278-8888
- Email: StudentITHelpDesk@fullerton.edu

For this course

Prerequisites

- CPSC 375, MATH 237 or CPSC 335;
 - MATH 338, PSYC 201, PUBH 349, ISDS 210 or EGEN 323
- Computer Science or Computer Engineering or Data Science major or minor

Course Catalog Description

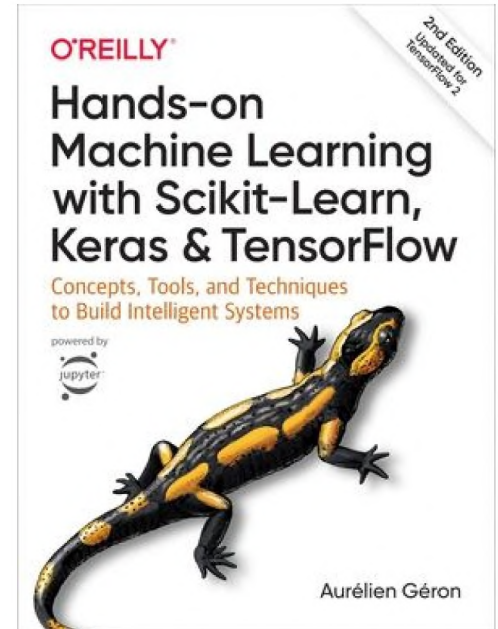
- Design, implement and analyze machine learning algorithms, including supervised learning and unsupervised learning algorithms. Methods to address uncertainty. Projects with real-world data.

Attendance: Not taken, but expected and strongly encouraged.

Textbook

Required Textbook: *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition*

Lecture notes will be uploaded within 24 to 48 hours after the class



Exams

- One midterm and one final exam (NOT cumulative)
 - Midterm: cover first half of the semester
 - Final: cover after the midterm to the end of the semester
- A **single double-sided cheat sheet** is allowed
 - No other external sources (e.g., cellphones, laptops, computers, etc.) nor collaboration
- Make-up exams will NOT be offered except in cases of extraordinary circumstances
 - Only if appropriate documentation can be provided & prior notice given ahead
- Requests for re-evaluation within one week
- PLEASE arrive on time!

Homeworks

- There will be 4 homework assignments
 - Due dates will be announced in classes and posted on Canvas
 - Collaboration is good but **NO PLAGIARISM**
- Questions or concerns regarding grading must be submitted in writing by email
 - Within one week of the grade being posted
- **Deadline:** All assignments must be submitted electronically via Canvas by the specified due date.
 - One-day grace period without penalty
 - Any work submitted after will NOT be accepted and will receive a grade of zero for that assignment

Group Projects

- One group project for groups of 1-5 people
- Approximately two submissions
 - First submission: project progress and division of responsibilities
 - Final submission: graded project submission
 - Only the **final submission** will be graded
- Final week reserved for **project presentations**

Detailed project overview and requirements next week

Survey paper

- **Required for graduate student**
 - Optional for undergraduate student (Extra Credit 5%)
- The topic must be directly related to the course content and approved in advance
- Submitted through Canvas no later than Friday of the final week at 11:59 p.m.
- IEEE two-column format in 11-point font,
 - Minimum length of **3 pages** (excluding references)

Grading Policies

Undergraduate

	<i>Percentage</i>
Homeworks	20%
Midterm Exam	25%
Final Exam	25%
Project	30%
Survey Paper (EC)	5%

Graduate

	<i>Percentage</i>
Homeworks	20%
Midterm Exam	25%
Final Exam	25%
Project	25%
Survey Paper	5%

Please read the syllabus carefully and discuss with me if necessary

Class Norm

- If I (or someone) use an odd word or a culturally-flavored example, let me know ASAP
- If you prefer to be called something other than your first name, please let me know
 - I'll make a best-effort to remember and adopt it
- I'll start class on time
- I'll aim to end class on time
 - Call me out if I go long
- Please mute your devices
- Any accommodations?
 - Glad to help!
 - Rather be involved sooner than later

Let's get started!

What is Machine Learning?

VS

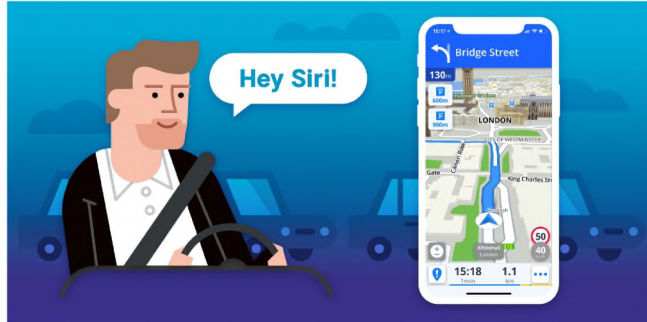
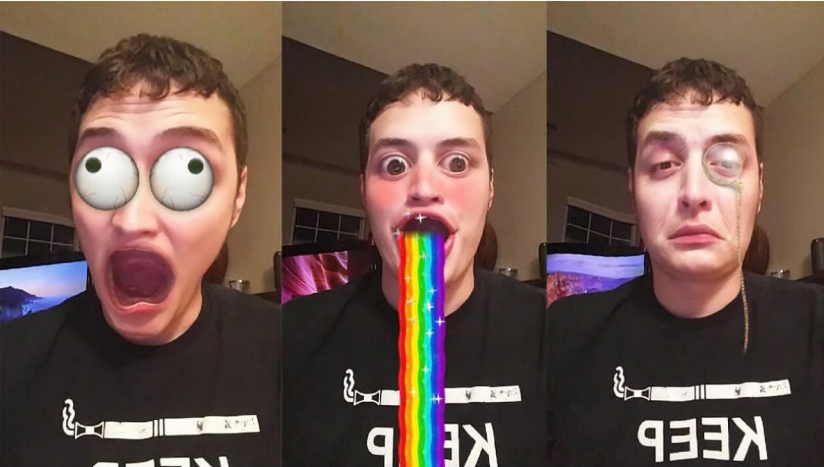
What is Artificial Intelligence?

Chapter 1

Machine Learning Landscape

- What is Machine Learning?
- Why Use Machine Learning?
- Examples of Applications
- Types of Machine Learning Systems
- Main Challenges of Machine Learning

AI?



Google

how do you fry an egg

X

🔊

🖼️

🔍

All

Videos

Forums

Images

Shopping

More

Tools

Without flipping

In a microwave

In water

Sunny side up

Over medium

How hot

About 278,000,000 results (0.31 seconds)

AI overviews are experimental. [Learn more](#)

To fry an egg, you can follow these steps:

1. Heat butter or oil in a non-stick skillet over medium or medium-low heat
2. Crack the egg into the pan
3. Cook until the whites are set, which can take about 3 minutes
4. Flip the egg and cook until the yolk reaches your desired doneness, which can take about 2–3 minutes for over easy, or longer for over medium or hard
5. Season with salt and pepper



Budget Bytes

How to Fry an Egg - Budget Bytes

May 25, 2022



Delish

How to Fry an Egg - Perfect Fried Egg Over Easy, Medium & Hard...

Apr 5, 2023 — Step 1In a small nonstick over medium heat, melt butter (or...



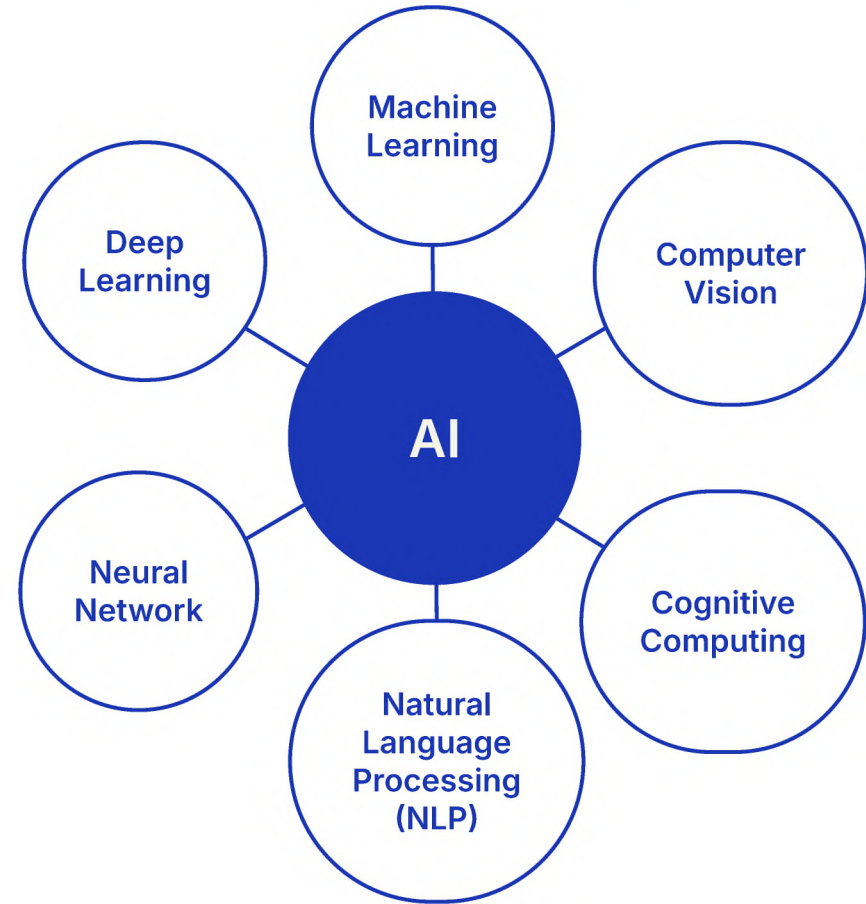
American Egg

How to Fry an E Incredible Egg - Directions, For Ov Hard Eggs: HEAT ;

What is Artificial Intelligence?

“Predicting the future isn’t magic, it’s artificial intelligence.” -Dave Waters

- The **simulation of human intelligence processes by machines**, enabling them to perform tasks that typically require human cognitive functions



What is Learning?

- **Learning** is a process of **acquiring knowledge**
 - Knowledge: Facts (data), patterns, concepts, rules, models, skills...

Example forms of knowledge

- Facts: A thing that is known to be true
 - e.g., a red car, $\pi=3.141592\dots$
- Patterns: repeated forms that represent the nature of facts
- Concepts: classes, class hierarchies created by generalization from patterns
 - e.g., 0, 2, 4, 6, ... $\Rightarrow 2n$ (even number concept)
 - Dog, Cat, Bird $\Rightarrow$ Animal (animal class hierarchy)
- Rules: $P \rightarrow Q$
 - e.g., if it rains, ground is wet.
 - If oil price is too high, people travel less
- Models: Abstract representation
 - e.g., mathematical functions such as $v = d/t$, $f(n) = f(n-1) + f(n-2)$, data model, design model, simulated systems
- Skills: Being able to do a complex activities,
 - e.g., playing soccer, writing programs, etc

Definition of Machine Learning

- “A field of study that gives computers the **ability to learn** without being explicitly programmed.”

-Arthur Samuel, 1959

- “Any change in a system that allows it to **perform better** the second time on repetition of the same task or on another task drawn from the same population.”

-Herbert Simon, 1983

- “A computer program that learns from **experience E** with respect to some **class of tasks T** and **performance measure P**. If its performance at tasks in T, is measured by P, it improves with experience E.”

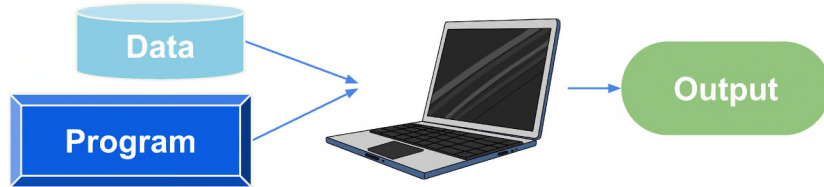
-Tom Mitchell, 1997

ML is the science/art of programming computers so they can **learn from data**

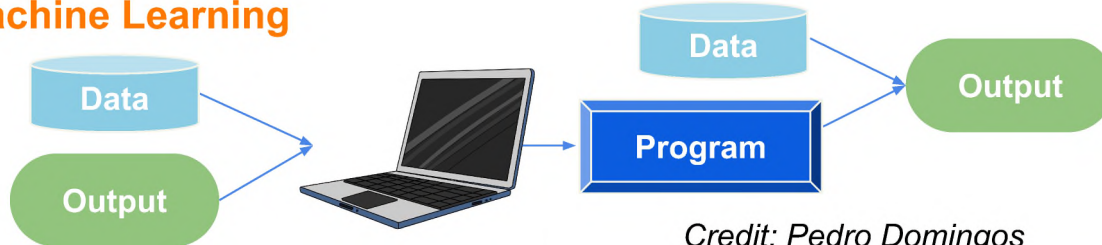
Why Machine Learning Matters?

- Data explosion
- Real-time decision-making needs
- Shift from rule-based systems to data-driven systems

Traditional Programming



Machine Learning



Credit: Pedro Domingos

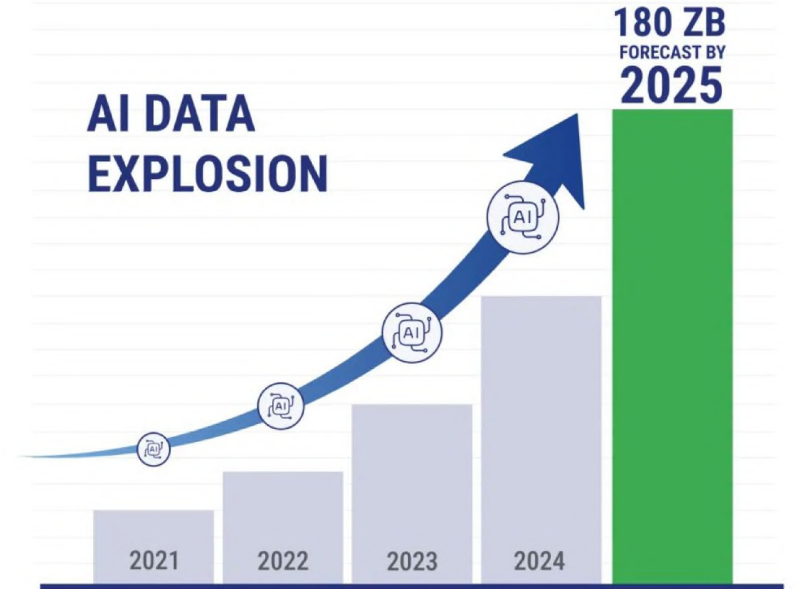
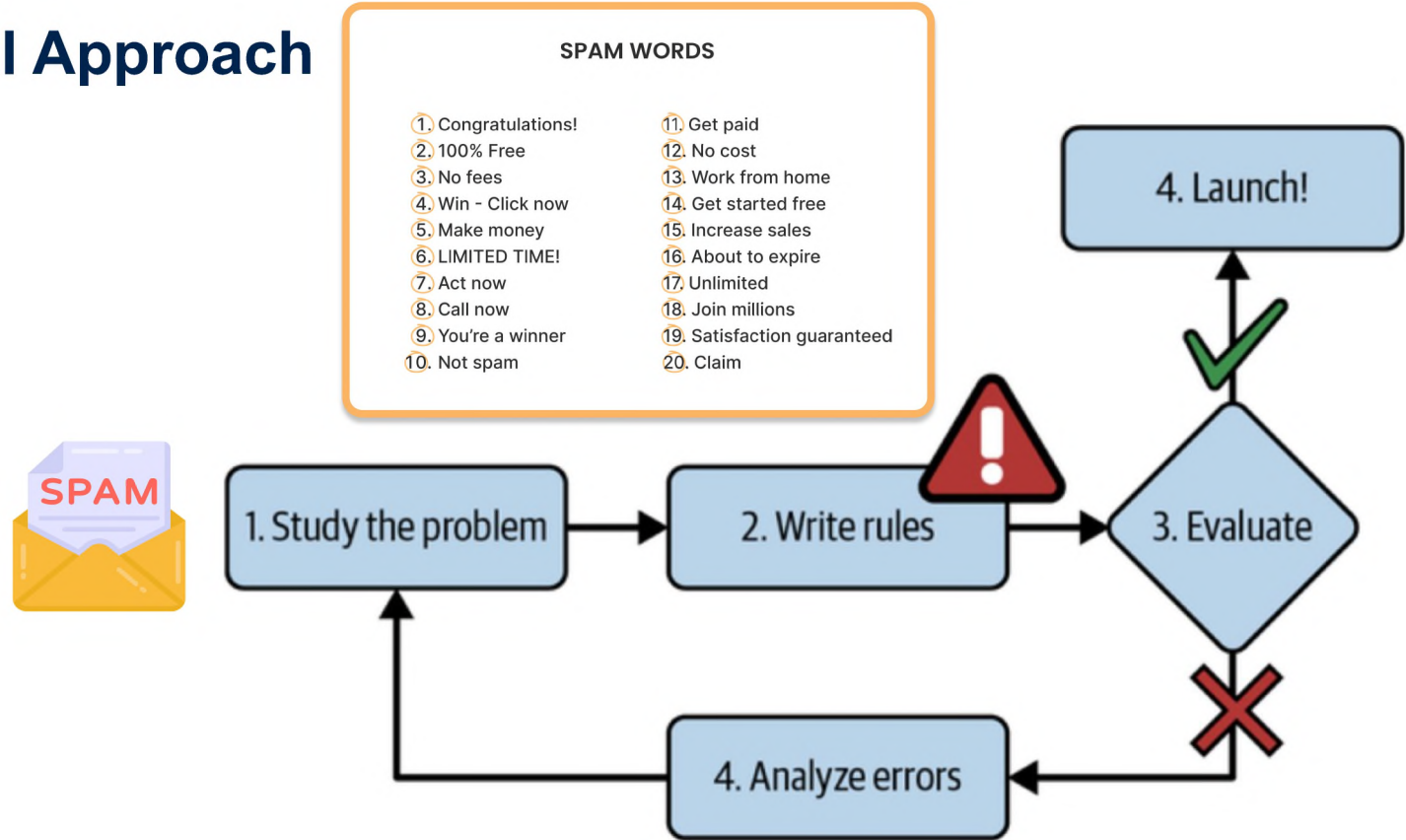
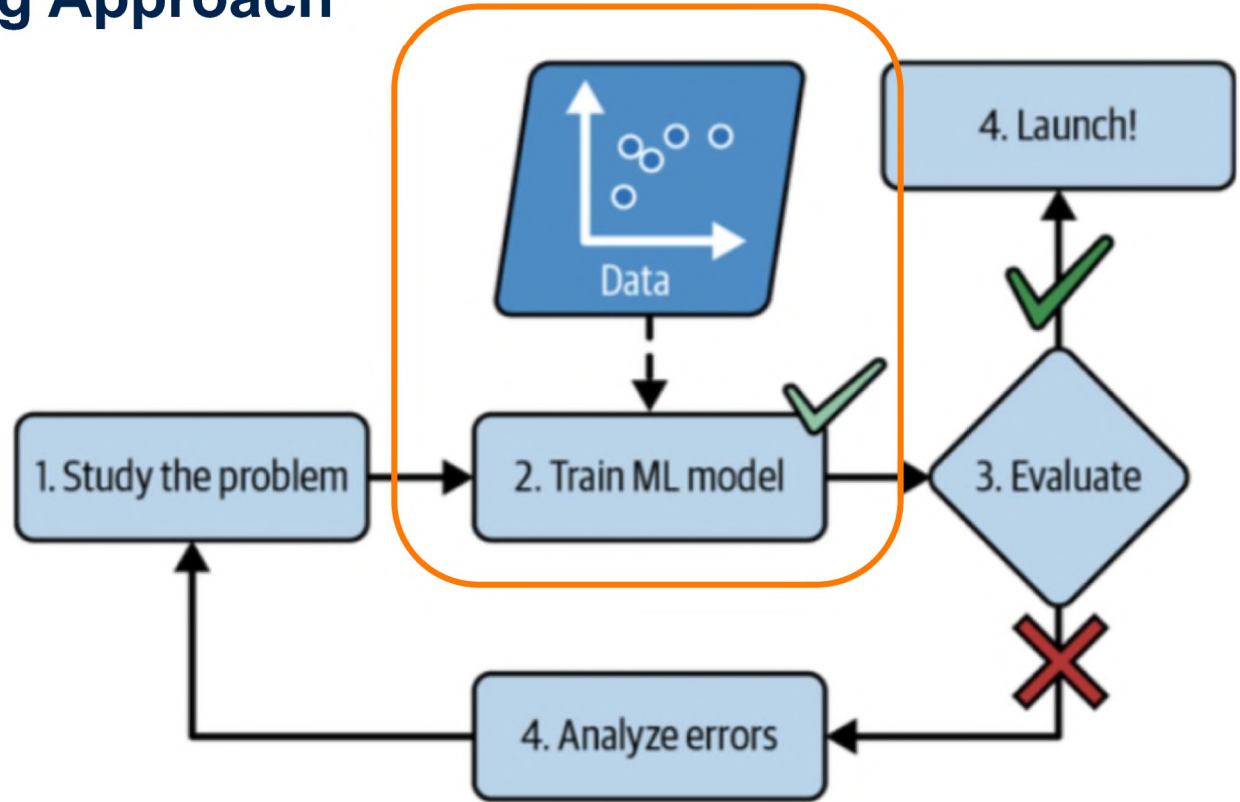


Image courtesy: Matt Pacheco 2024

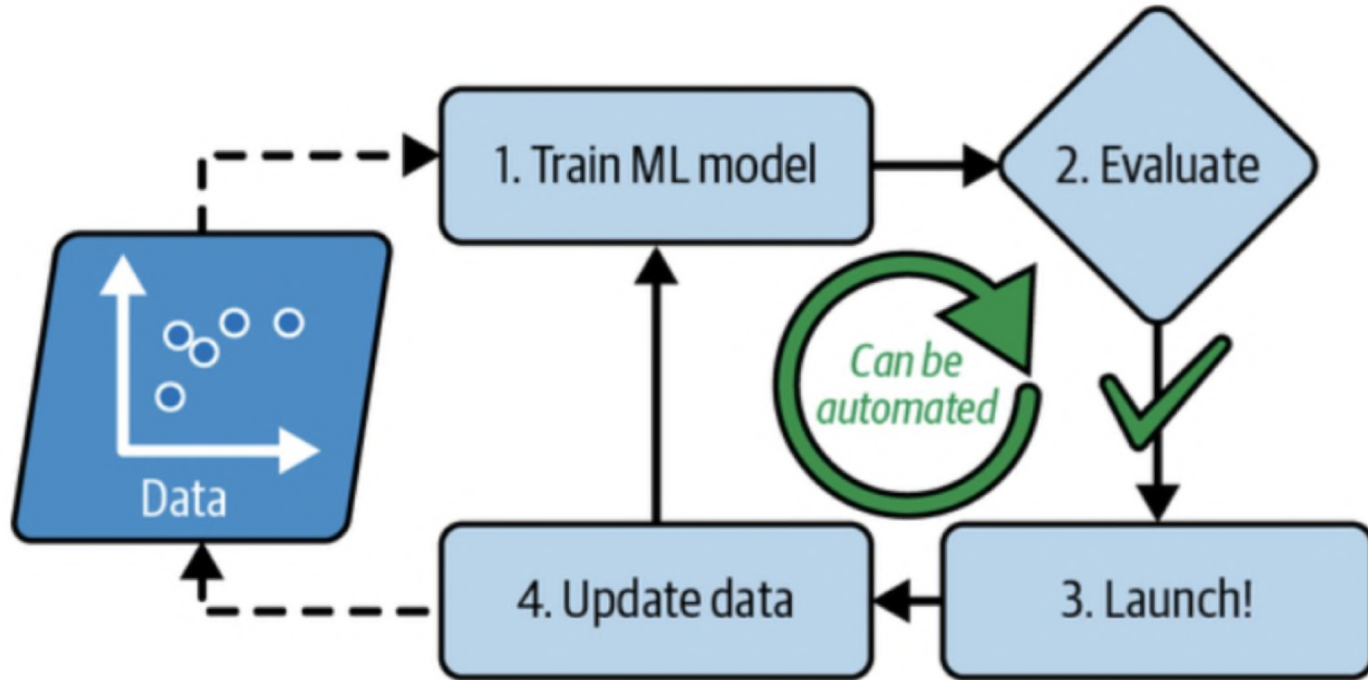
Traditional Approach



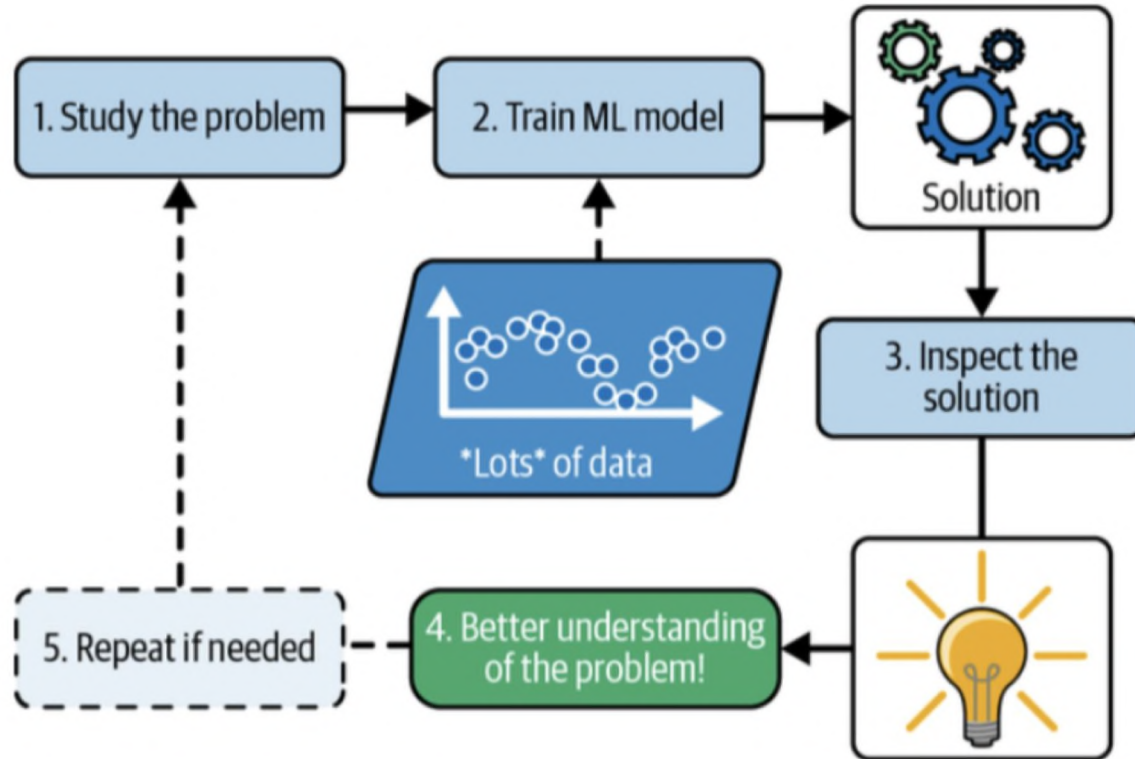
Machine Learning Approach



Benefit: Automatically adapting to change

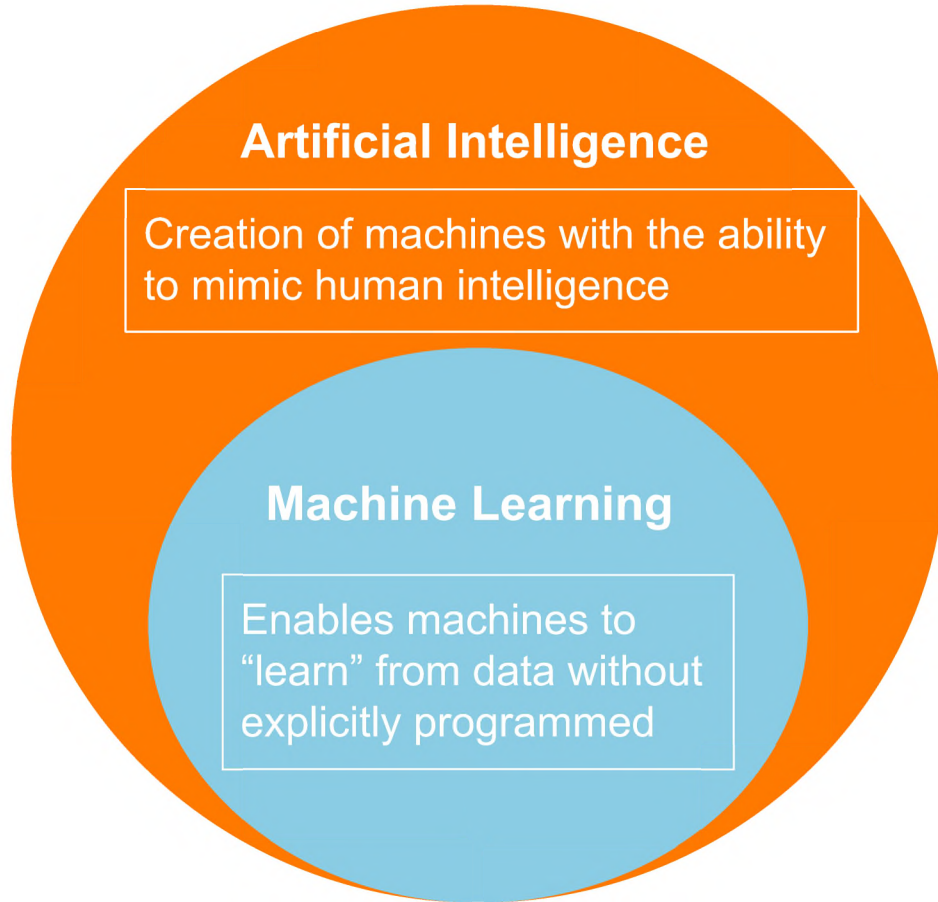


Benefit: ML can help humans understand a problem



Why Machine Learning?

- Problems for which existing solutions require a lot of fine-tuning or long lists of rules
- Fluctuating environments
 - a ML system can easily be retrained on new data, always keeping it up to date
- Getting insights about complex problems and large amounts of data
- Complex problems for which using a traditional approach yields no good solution



Classical Examples of ML Tasks

Pattern Recognition

- faces, speech, handwriting...



Image courtesy: S. Mattoo 2024; F. Ildillah 2018



Classical Examples of ML Tasks

Pattern Recognition

- faces, speech, handwriting...

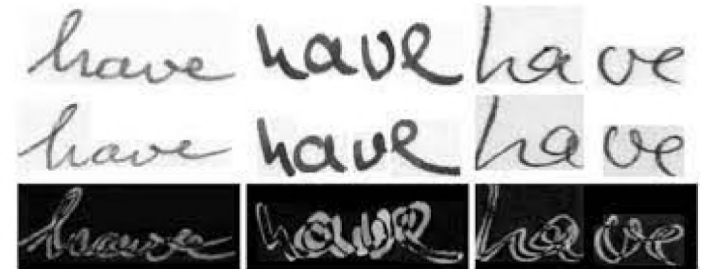
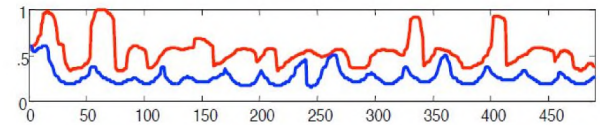
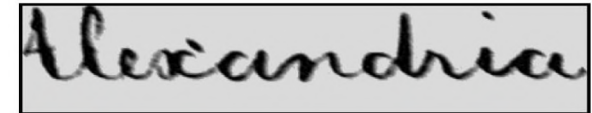
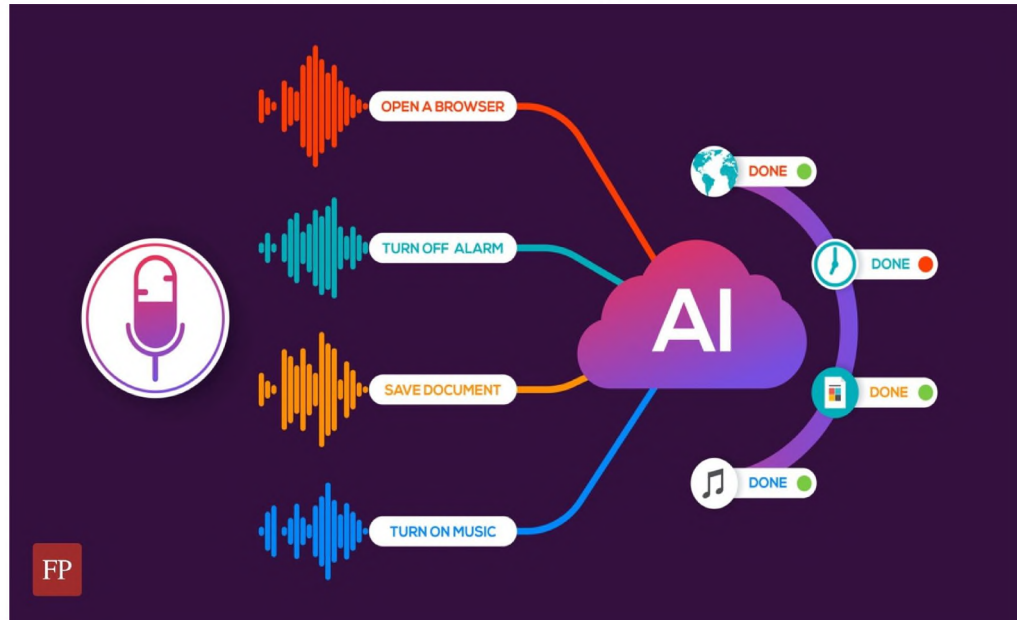


Image courtesy: M. Hanny Sabbagh; G. Prokopiou 2024; S. Xiao 2020

Classical Examples of ML Tasks

Pattern Generation

- motion, image synthesis...

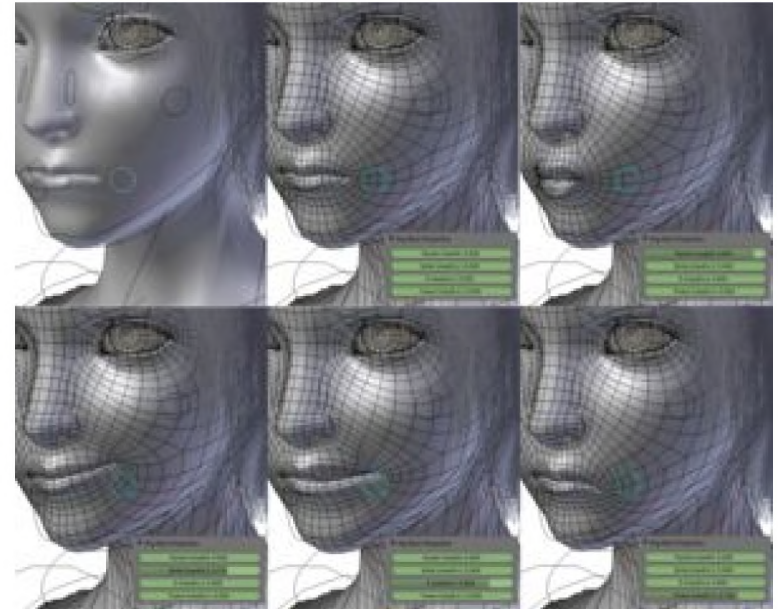
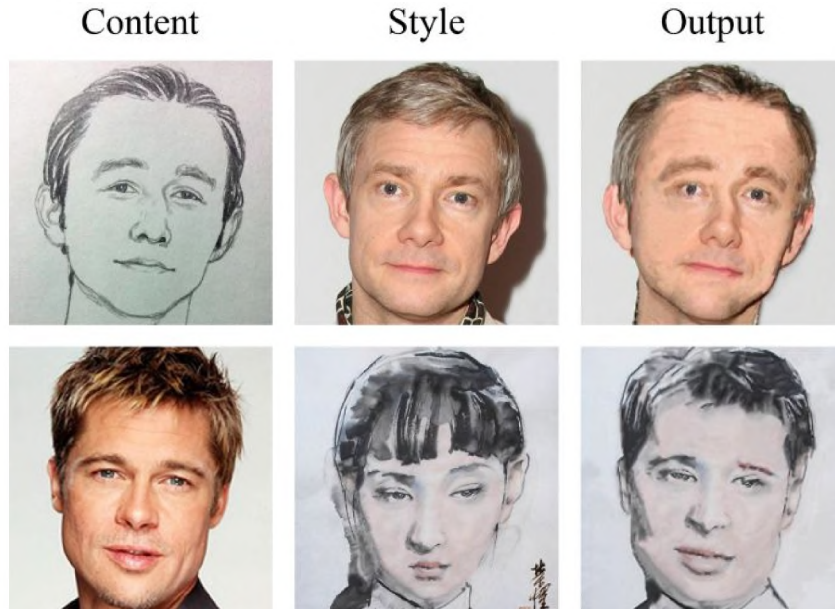


Image courtesy: M. Abdollahnejad 2020; M. Hanny Sabbagh

Classical Examples of ML Tasks

Anomaly Detection

- fraud, unusual sensor reading...

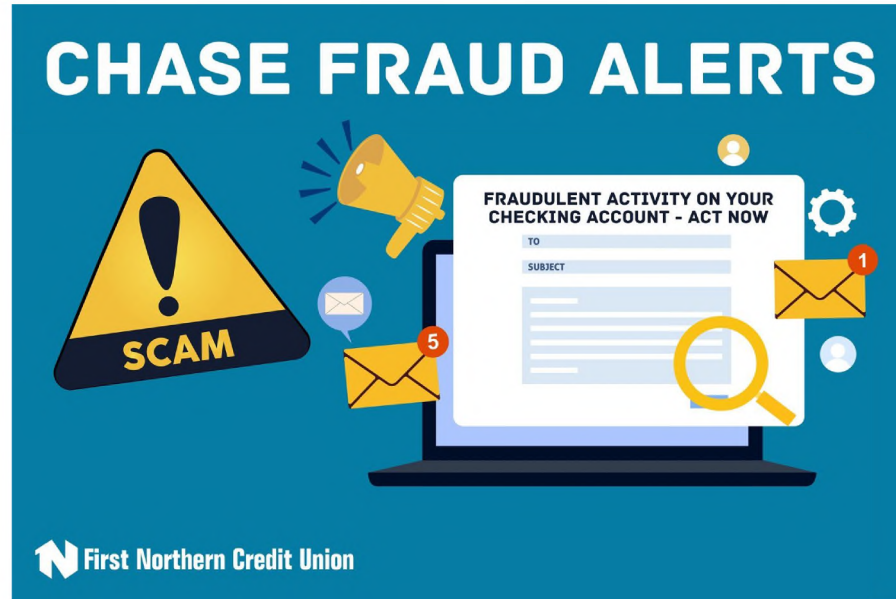
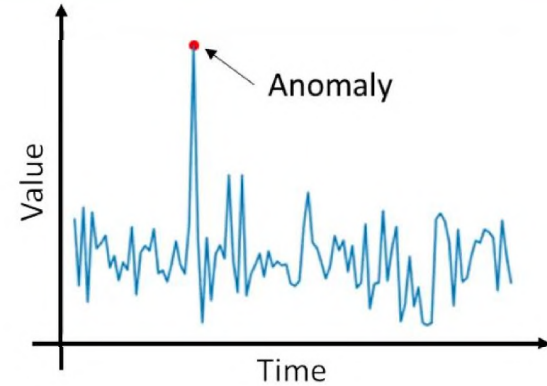


Image courtesy: Aura; Google



Chase Fraud: Did you use card ending [redacted] for \$0.02 at [basilplay.com](https://www.basilplay.com) on 04/ ? Reply YES or NO. Msgs&rates may apply.

No

Chase Thanks. Your card can't be used until we speak to you. Call [800-454-9078](tel:8004549078) or the number on your card to turn it on or order a new one.

Classical Examples of ML Tasks

Prediction

- market trends, health, weather...

Apple Inc

\$207.23 ↓ 6.46% -14.31 5D

Pre-market: \$207.90 (↑ 0.32%) +0.67

Closed: Aug 7, 9:19:20 AM UTC-4 · USD · NASDAQ · Disclaimer

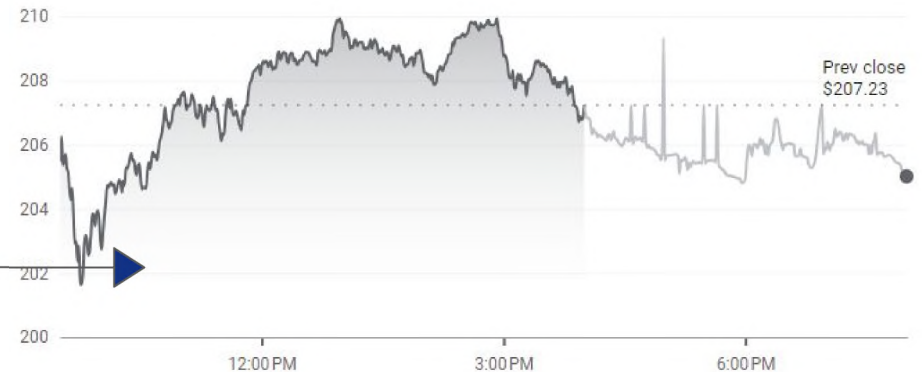


Image courtesy: Google Finance

Classical Examples of ML Tasks

Prediction

- market trends, health, weather...

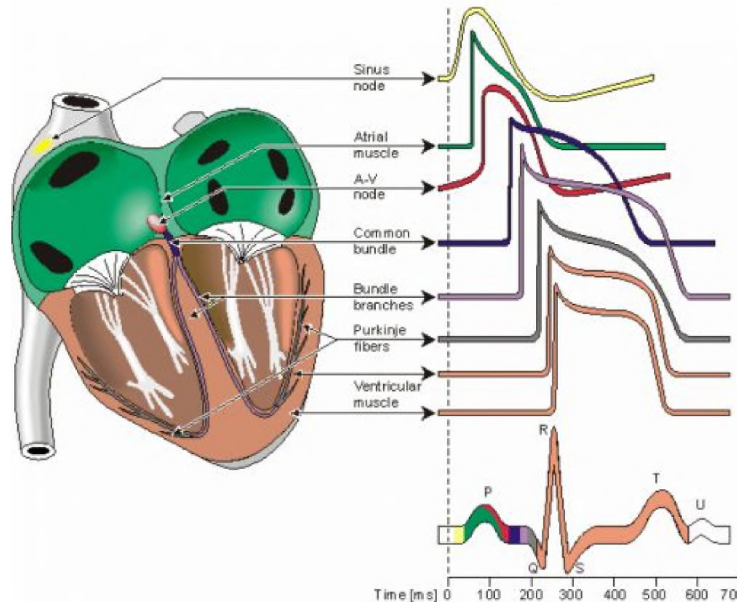


Image courtesy: B. Cornell 2016; Kubios

Normal Sinus Rhythm



Complexes normal, evenly spaced. Rate 60–100 bpm

Bradycardia



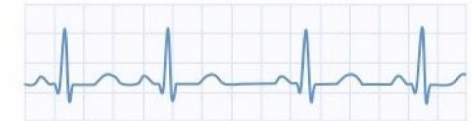
Complexes normal, evenly spaced. Rate < 60 bpm

Atrial Fibrillation



Baseline irregular. Ventricular response irregular

Arrhythmia



All complexes normal, rhythm irregular

Tachycardia



Complexes normal, evenly spaced. Rate > 100 bpm

Ventricular Fibrillation



Rapid, wide irregular ventricular complexes

Application: Autonomous Cars

Self-Driving Car

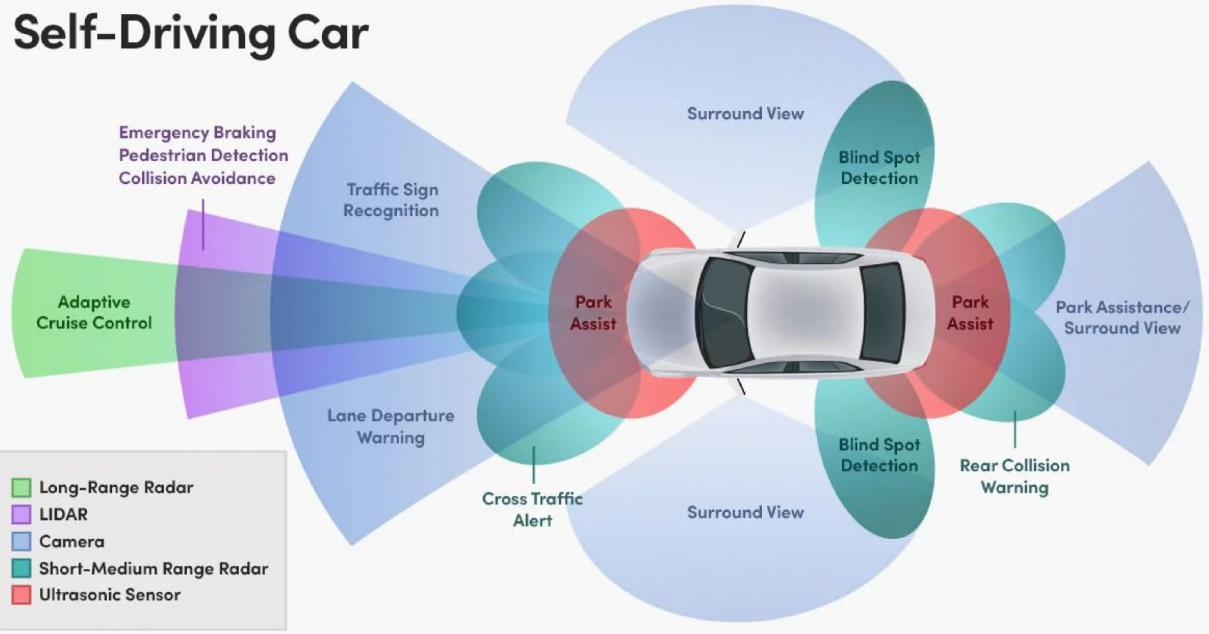
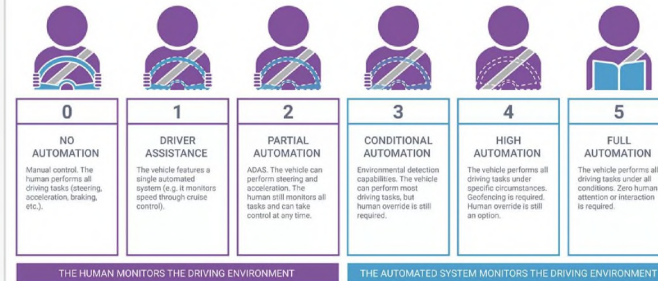


Image courtesy: Paul Christianson 2020

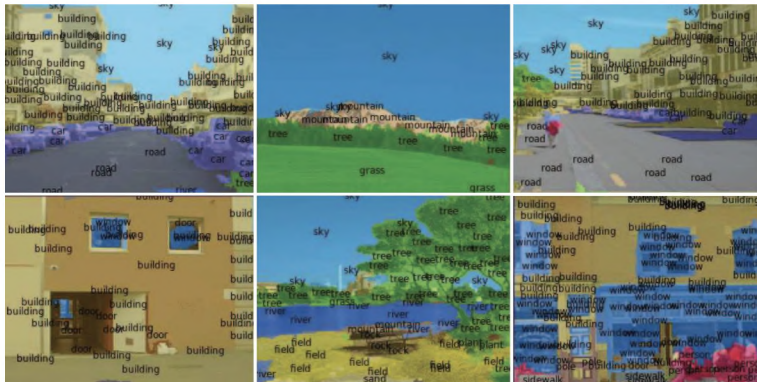
Image courtesy: Melissa Kirschner 2019

synopsys®

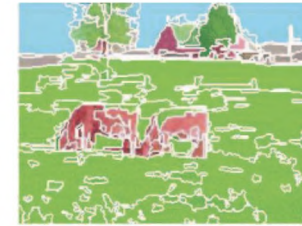
LEVELS OF DRIVING AUTOMATION



Application: Scene Labeling



(a) Image



(b) Superpixels



(c) Groundtruth



(d) Threshold in gPb hierarchy



(e) CRF in gPb hierarchy



(f) MinCover in gPb hierarchy

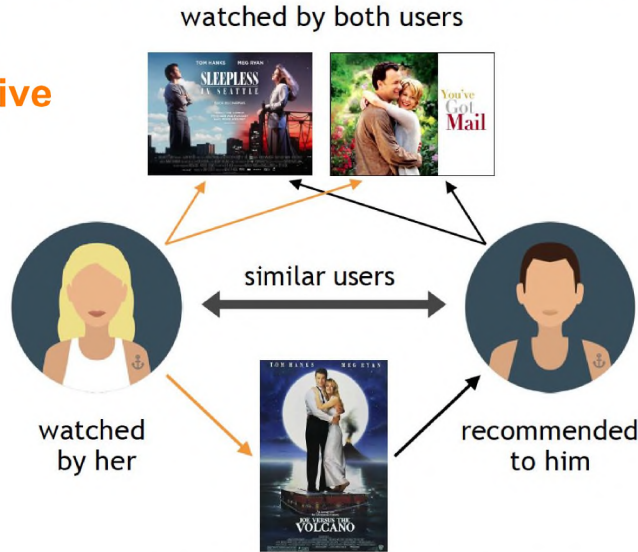


Legend: Building Sky Grass Mountain Tree Object

Image courtesy: Clément Farabet et al. 2012

Application: Recommendation System

Collaborative



Content-based

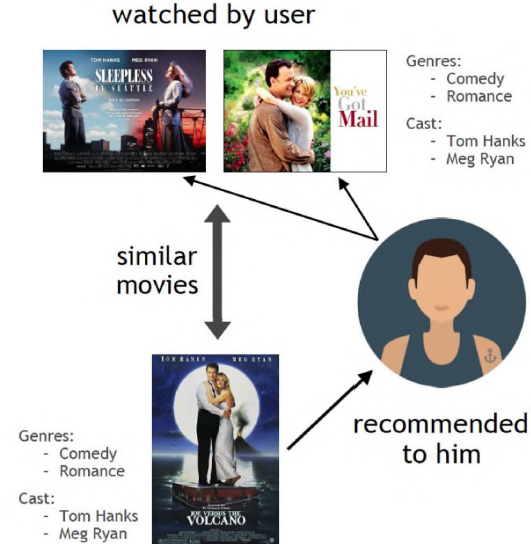


Image courtesy:
NVIDIA; Spotify

MORE LIKE
Billie Eilish

American Teen
Khalid

Reflection
Fifth Harmony

Lo Vas A Olvidar
Rosalía

Made For You

Discover Weekly
Enjoy new discoveries
chosen just for you!

Your Daily Drive
Fans like you also
like these songs!

Main Challenges of Machine Learning

Machine learning is **ill-posed**

- limited sample of the world/real problem

NEW GOAL: a model that **generalizes** beyond the dataset and that is not influenced by the noise in the dataset

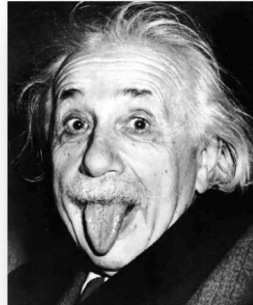


- **UPPER BOUND/Occam's Razor:** "With all things being equal, the simplest explanation tends to be the right one."

← William of Ockham (1288-1348)

- **LOWER BOUND/Albert Einstein:** "Everything should be made as simple as possible, but not simpler."

Albert Einstein (1879-1955)



Scenarios

Students



A

B

C

Professor



Also available:

<https://www.analyticsvidhya.com/blog/2020/02/underfitting-overfitting-best-fitting-machine-learning/>

Scenarios



A

- Hobby = chating
- Not interested in class
- Doesn't pay much attention to professor

Scenarios



B

- Hobby = to be best in class.
- Mugs up everything professor says.
- Too much attention to the class work.

Scenarios



C

- Hobby = learning new things
- Eager to learn concepts.
- Pays attention to class and learns the idea behind solving a problem.

Scenarios

Students



A

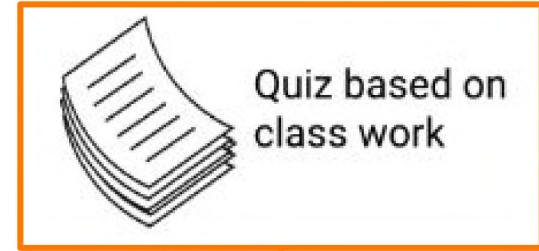


B



C

Professor



How would they do?

Scenarios



Guessing: ~50%

A



Mr. know it all
~98%

B



Problem solving approach:
~92%

C



Quiz based on
class work

Professor

Annie



Scenarios

Students



A

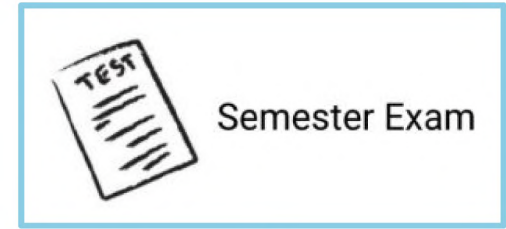


B



C

Professor



How would they do?

Scenarios



A

Not interested in learning

Class test ~50%

Test ~47%

Under-fit/ biased learning



B

Memorizing the lessons

Class test ~98%

Test ~69%

Over-fit/ Memorizing



C

Conceptual Learning

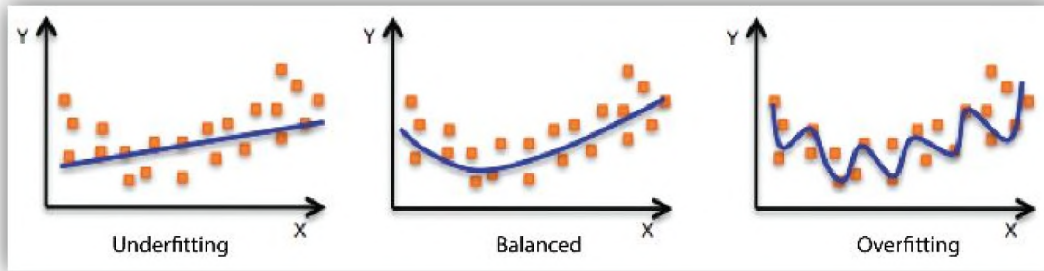
Class test ~92%

Test ~89%

Best-fit

Challenges - Overfitting & Underfitting

Image courtesy: bigdataspecialist

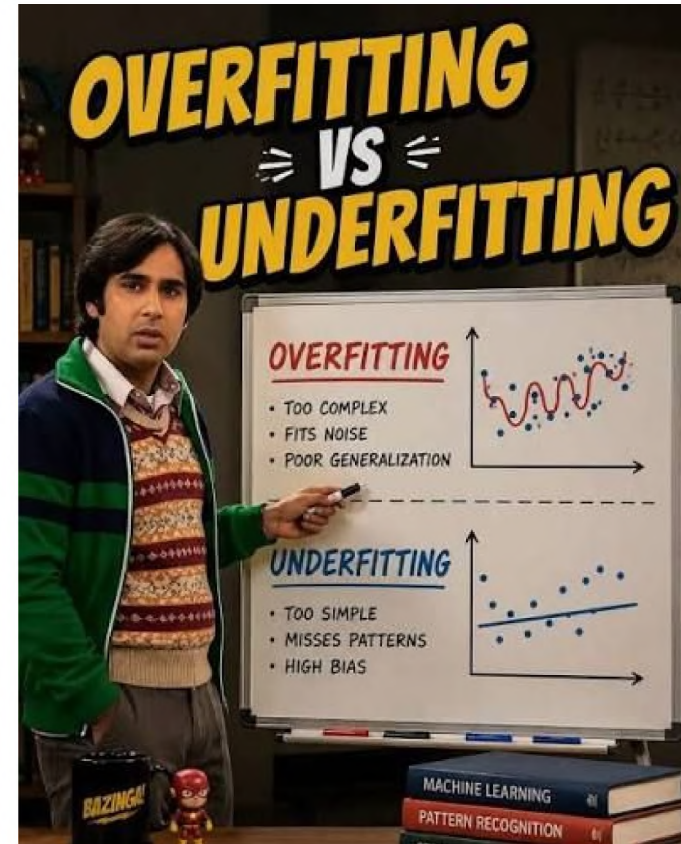


Underfitting

- Over simplified

Overfitting

- Over complicate



Criteria for Choosing Models

Inductive bias

- The set of assumptions that define the model selection criteria of a ML algorithm
 - Necessary for learning (beyond the dataset)
 - Otherwise we could only perform memorization (of our training set) without generalization capability

Usually two types of bias that we can use:

Restriction bias

- constrains the set of models that the algorithm will consider during the learning process.

Preference bias

- guides the learning algorithm to prefer certain models over others.

Types of Machine Learning

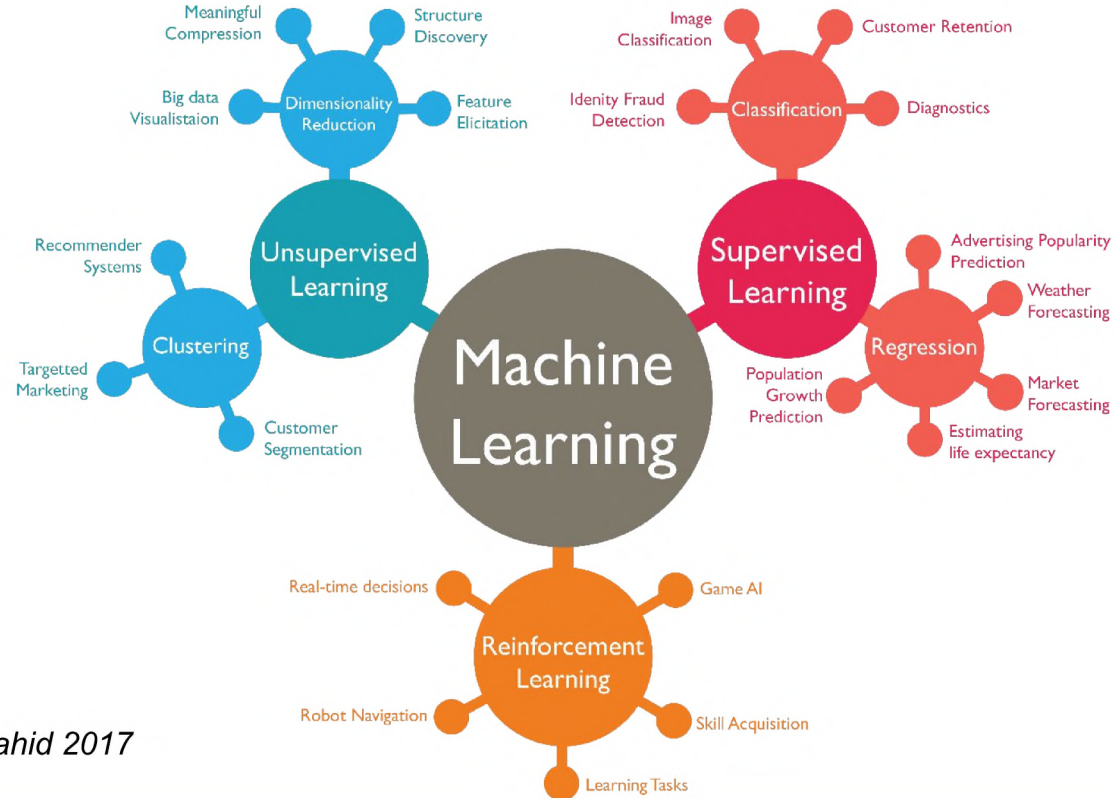


Image courtesy: Abdul Rahid 2017